

1. တွက်ချက်ရန်

$$A | \text{တံ 1} \quad m_1 = 0.6 \times 0.5 = 0.3$$

$$B | \text{တံ 2} \quad m_2 = 0.2 \times 0.5 = 0.1$$

$$\text{တစ်ခုခုရရှိရန် အတိအကျ အားကိုးမှု} = 0.3 + 0.1 = 0.4$$

$$\frac{0.1 \times 0.5}{0.4} = 0.125$$

$$0.4$$

$$m_1(\{T, B\}) = 0.5$$

$$m_1(\oplus) = 0.5$$

$$m_2(T) = 0.4$$

$$m_2(\oplus) = 0.2$$

| Intersection          | $m_1(\{T\}) = 0.4$ | $m_2(\oplus) = 0.2$ |
|-----------------------|--------------------|---------------------|
| $m_1(\{T, B\}) = 0.5$ | $ST = 0.4$         | $\{T, B\} = 0.1$    |
| $m_1(\oplus) = 0.5$   | $ST = 0.4$         | $\oplus = 0.1$      |

$$m_3(\{T\}) = 0.4$$

$$m_3(\{T, B\}) = 0.1$$

$$m_3(\oplus) = 0.1$$

2. Decision Tree

$$E(S) = -(\text{True/ရရှိမှု}) \log_2(\text{True/ရရှိမှု}) - (\text{False/ရရှိမှု}) \log_2(\text{False/ရရှိမှု})$$

$$IG(Att) = E(S) - E(Att)$$

IG အကန့်အသတ် Root

7. วิจัยหาความถี่ของยีนและอัลลีลในประชากร  $y = 4x^2 + 3$

1. x สุ่ม 0 ถึง 15

2. ขนาดประชากร 5

3. โยธินทรีย์ 1 ชนิด

4. Crossover rate:  $P_c = 0.6$

5. ฟังก์ชัน  $y = 4x^2 + 3$

6. อัตราการกลายพันธุ์ Mutation Rate:  $P_m = 0.01$

7. ความยาวโครโมโซม  $y = 4x^2 + 3$

8. สุ่มค่า 30 ครั้ง

9. คำนวณค่าเฉลี่ยของโครโมโซม โดยวิธี Monte Carlo

10. เปรียบเทียบผล

| * ขนาดประชากร 5 | ค่าสุ่ม | โครโมโซม | x  | $y = 4x^2 + 3$ |
|-----------------|---------|----------|----|----------------|
| 0-15            | 1       | 0010     | 2  | 19             |
| 4 bit           | 2       | 1000     | 8  | 259            |
|                 | 3       | 1111     | 15 | 903            |
|                 | 4       | 1010     | 10 | 403            |
|                 | 5       | 0100     | 4  | 67             |

$$p_{select} = \frac{F_i}{\sum F}$$

$$E_i = p_{select} \times \text{size} \quad \text{จำนวนครั้งที่เลือก}$$

| ลำดับ | โครโมโซม | x  | F, y | $p_{select}$ | $E_i$ |   |
|-------|----------|----|------|--------------|-------|---|
| 1     | 0010     | 2  | 19   | 0.011        | 0.055 | 0 |
| 2     | 1000     | 8  | 259  | 0.157        | 0.785 | 1 |
| 3     | 1111     | 15 | 903  | 0.547        | 2.735 | 4 |
| 4     | 1010     | 10 | 403  | 0.244        | 1.22  | 0 |
| 5     | 0100     | 4  | 67   | 0.041        | 0.205 | 0 |

| ลำดับ                     | 1     | 2     | 3     | 4     | 5     |
|---------------------------|-------|-------|-------|-------|-------|
| F                         | 19    | 259   | 903   | 403   | 67    |
| $p_{select}$              | 0.011 | 0.157 | 0.547 | 0.244 | 0.041 |
| $q_i$                     | 0.011 | 0.169 | 0.915 | 0.959 | 1.00  |
| $r$                       | 0.199 | 0.444 | 0.915 | 0.126 | 0.777 |
| $q_{i-1} \leq r \leq q_i$ | 2     | 3     | 3     | 1     | 3     |

$$P_c \leq 0.6$$

|        |        | Date           |            | No.    |                            |
|--------|--------|----------------|------------|--------|----------------------------|
| លំដាប់ | លំដាប់ | កំរិត $\gamma$ | លំដាប់     | លំដាប់ | $\times$                   |
| 3      | 1111   | 2,3            | 0.59       | 1000   | <del>3</del> 1001 9 329 1  |
| 3      | 1111   |                | $\leq 0.6$ | 1111   | <del>3</del> 1110 14 787 2 |
| 3      | 1111   | 3,3            |            |        | 1111 15 903 3              |
| 2      | 1000   |                |            |        | 1111 15 903 4              |
| 3      | 1111   | 3,3            |            |        | 1111 15 903 5              |

| លំដាប់ | លំដាប់ | កំរិត ( $\gamma$ ) | លំដាប់            |
|--------|--------|--------------------|-------------------|
| 0.09   | 1001   | 0.01               | 0.7 0.11 0.65     |
|        | 1110   | 0.17               | 0.90 0.97 0.18    |
|        | 1111   | 0.047              | 0.646 0.491 0.029 |
|        | 1111   | 0.653              | 0.123 0.915 0.171 |
|        | 1111   | 0.175              | 0.009 0.579 0.062 |

|                                                               | NOR | A | B | A+B |
|---------------------------------------------------------------|-----|---|---|-----|
| $w_1 = 1.80$ Bias = 1                                         | 0   | 0 | 1 | /   |
| $w_2 = 1.10$ $\theta = \text{Bias} \times w_3 = 1 \times 0.5$ | 0   | 1 | 0 | /   |
| $w_3 = 0.50$ $\eta = 0.9$                                     | 1   | 0 | 0 |     |
| 1 epoch                                                       | 1   | 1 | 0 |     |

$$\begin{aligned}
 \text{net} &= A W_1 + B W_2 + \text{bias } W_3 \\
 \leq 0 & \text{ out} = f_{\text{net}} = f_n(0.5) = 0 \\
 & W_1 = 1.80 + 0.9(1-0)0^A = 1.8 \\
 & W_2 = 1.10 + 0.9(1-0)0^B = 1.1 \\
 & W_3 = 0.5 + 0.9(1-0)1^{\text{bias}} = 1.2 \\
 \text{err} &= d - \text{out} = 1 - 0 = 1
 \end{aligned}$$

(2)

|       |                             |         |
|-------|-----------------------------|---------|
| net   | $= 0(19) + 1(1.1) + 1(1.2)$ | $= 2.3$ |
| out   | $= f(2.3)$                  | $= 1$   |
| $W_1$ | $= 19 + 0.9(0 - 1)0$        | $= 19$  |
| $W_2$ | $= 1.1 + 0.9(0 - 1)1$       | $= 0.4$ |
| $W_3$ | $= 1.2 + 0.9(0 - 1)1$       | $= 0.5$ |
| err   | $=  0 - 1 $                 | $= 1$   |

③

(A)

DL 1 epoch

# Sigmoid

3 98 NN XOR

$$x_1 = 0, x_2 = 1, y_{d,5} = 1, \alpha = 0.4, W_{13} = 0.37$$

$$W_{14} = 0.74, W_{23} = 0.56, W_{35} = 0.71, W_{45} = 1.35, \theta_3 = 0.92$$

$$\theta_4 = -0.55, \theta_5 = 0.46, W_{24} = -1.04$$

$$y_3 = 1/[1 + e^{-(x_1 W_{13} + x_2 W_{23} - \theta_3)}] = 0.435$$

$$y_4 = 1/[1 + e^{-(x_1 W_{14} + x_2 W_{24} - \theta_4)}] = 0.380$$

$$y_5 = 1/[1 + e^{-(y_3 W_{35} + y_4 W_{45} - \theta_5)}] = 0.589$$

ค่าผิดพลาด

$$e = y_{d,5} - y_5 = 0.411$$

$$\delta_5 = y_5(1 - y_5)e = 0.099$$

$$\Delta W_{35} = \alpha \times y_3 \times \delta_5 = 0.0172$$

$$\Delta W_{45} = \alpha \times y_4 \times \delta_5 = 0.0150$$

$$\Delta \theta_5 = \alpha \times (-1) \times \delta_5 = -0.0396$$

error layer 3, 4

$$\delta_3 = y_3(1 - y_3) \times \delta_5 \times W_{35} = 0.00040$$

$$\delta_4 = y_4(1 - y_4) \times \delta_5 \times W_{45} = 0.00035$$

การปรับน้ำหนัก

$$\Delta W_{13} = \alpha \times x_1 \times \delta_3 = 0$$

$$\Delta W_{23} = \alpha \times x_2 \times \delta_3 = 0.00016$$

$$\Delta \theta_3 = \alpha \times (-1) \times \delta_3 = -0.00016$$

$$\Delta W_{14} = \alpha \times x_1 \times \delta_4 = 0$$

$$\Delta W_{24} = \alpha \times x_2 \times \delta_4 = 0.00014$$

$$\Delta \theta_4 = \alpha \times (-1) \times \delta_4 = -0.00014$$

$$W_{13} = W_{13} + \Delta W_{13} = 0.37, W_{45} = W_{45} + \Delta W_{45} = 1.365$$

$$W_{14} = W_{14} + \Delta W_{14} = 0.74, \theta_3 = \theta_3 + \Delta \theta_3 = 0.91984$$

$$W_{23} = W_{23} + \Delta W_{23} = 0.56016, \theta_4 = \theta_4 + \Delta \theta_4 = -0.55014$$

$$W_{24} = W_{24} + \Delta W_{24} = -1.03986, \theta_5 = \theta_5 + \Delta \theta_5 = 0.4204$$

$$W_{35} = W_{35} + \Delta W_{35} = 0.7272$$